Informational Energetics: A Universal Architecture of Persistence

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Any system that persists must solve a universal control problem. We present Informational Energetics, a framework modeling the persistence of complex systems against entropy, and validate it by deriving the fine-structure constant from first principles. By reinterpreting robust control theory through information theory and thermodynamics, we synthesize the Universal Architecture of Persistence: a recursive six-pillar structure (Capacity, Map, Protocol, Governor, Toll, and Margin). To validate the framework, we apply it to the vacuum. While standard physics parameterizes vacuum properties through measurement, we show that these same parameters emerge as the unique geometric solutions to a persistence problem: maintaining coherence with zero external energy. This paper establishes the theoretical foundation for the E_8-Persistence Theory by deriving its core architectural requirements: Finiteness, Unitarity, and Causality. These constraints mandate a substrate that is positive-definite, even, and self-dual; an E_8 lattice projected onto $D = 4$ spacetime is the unique geometric substrate solution. This geometry is characterized by the derived invariant integers $\mathbb{S} = \{\Delta{=}43 \text{ (temporal)}, \nu{=}16 \text{ (chiral)}, \sigma{=}5 \text{ (interaction)}, \chi{=}2 \text{ (topological)}\}$. The theory makes a falsifiable prediction for all physical constants with no continuous free parameters; all coefficients are fixed by the characteristic integers of the substrate. We validate this by deriving the Fine-Structure Constant (α^{-1}) as the Geometric Impedance of the substrate:

$$\alpha^{-1} = \underbrace{\pi\Delta}_{\text{Capacity}} + \underbrace{\chi}_{\text{Map}} - \underbrace{\frac{1}{R_M - \sigma}}_{\text{Protocol}} - \underbrace{\frac{\chi}{\Delta}}_{\text{Governor}} + \underbrace{\frac{1 \cdot \chi \cdot (R_M - \sigma)}{N^3 \cdot \sigma \cdot R_M}}_{\text{Toll}} + \underbrace{\frac{1}{L_{embed} \cdot (\sigma + 1) \cdot \Delta^2}}_{\text{Margin}}$$

where $R_M = D\Delta$ is manifold resolution, $N = 2 \cdot \nu$ is node capacity, and $L_{embed} = \nu + \sigma + \chi + 2D$ is embedding load. This calculation yields $\alpha^{-1} = 137.035999212\ldots$, agreeing with Morel (2020) within 0.58σ and CODATA (2022) within 1.68σ. The Von Klitzing Constant (R_K) follow as geometric consequences.

I. INTRODUCTION

The structures of the observable universe, from biological cells to the quantum vacuum, share a defining property: they *persist* against the thermodynamic current that erases distinction. Control theory describes how systems regulate against fluctuations, while treating the regulation target (setpoint) as an externally specified boundary condition. Information theory quantifies the cost of information processing, but treats the processor as given. Thermodynamics establishes the arrow of time, but treats entropy as a property of *states*, not of *processing*.

Despite their individual successes, these three fields remain structurally separated. Control theory cannot explain where setpoints come from; information theory cannot explain where processors come from; thermodynamics establishes the tendency toward maximum entropy, not the conditions under which complexity survives. No existing framework treats persistence as the primary phenomenon to be explained, rather than as a background assumption. This separation reflects a missing theoretical layer.

To model persistence, in section II we synthesize **Informational Energetics** (IE): the study of how systems convert energy into structural information to resist entropic decay. This yields the **Universal Architecture of Persistence**: six existential constraints (Capacity, Map, Protocol, Governor, Toll, Margin) that define the lifetime of any persistent entity.

This paper validates IE against the vacuum, the minimal persistent substrate: it persists with zero external flux, has no deeper system beneath it, and admits no external setpoint. These requirements create the most constrained test case possible. If persistence requirements alone cannot derive the vacuum's structure, the framework is not universal.

Standard models of particle physics treat the dimensions, symmetries, and constants of the vacuum as empirically measured parameters. In this work, by re-framing the vacuum as an information-processing substrate, we derive these foundational parameters as emergent, optimal solutions to a systemic control problem aimed at maintaining structural coherence with zero external energy flux.

A. Information Theoretical Context of Physics

The concept that physical reality is fundamentally constrained by information processing costs is rooted in Landauer's principle [1] and discussed by Wheeler ("It from Bit")[2]. More recently, Verlinde[3] proposed that gravity

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is an entropic phenomenon emerging from information gradients and Bianconi[4] proposed that gravity arises from quantum entropy coupling matter with geometry. While concordant with Landauer, IE applies this logic broadly to all persistent systems, treating the minimization of Entropic Action as the primary driver of their dynamics. All derivations follow from applying this principle.

B. Testing the Framework: Deriving the Vacuum

We define the vacuum as the minimal persistent substrate from which all observable structures emerge. Because it admits no deeper persistent system beneath it, it cannot obtain configuration information, boundary conditions, or memory from an external source; this makes it the most constrained possible test case for the framework. We apply IE in three strict derivation layers, where each builds only upon the previous, with no appeal to phenomena not yet derived.

1. **System 0: Requirements.** We map the six pillars to physics, deriving Finiteness, Unitarity, and Causality as existential necessities (Section III).
2. **System 1: Substrate.** System 0's requirements uniquely specify an E_8 lattice projected to $D = 4$ spacetime, yielding the characteristic integers $\mathbb{S} = \{\Delta = 43, \nu = 16, \sigma = 5, \chi = 2\}$ (Section IV).
3. **System 2: Impedance.** We derive the fine-structure constant (α^{-1}) as a geometric consequence of this substrate (Section V). We further identify a predicted *Quantization Gap* between kinematic and high-loop QED extractions, consistent with the substrate's finite channel capacity, and propose a falsification criterion.

We stress that α^{-1} is constrained through an eliminative filter: IE, the definition of the substrate, the substrate geometry, and the characteristic integers jointly select a unique value with no continuous free parameters; all coefficients are fixed by the characteristic integers of the substrate.

1. The E_8 Lattice: Substrate vs. Algebra

The exceptional Lie group E_8 has long been explored as a candidate for unification because it is the smallest simple Lie group large enough to contain the Standard Model gauge group. Most famously, Garrett Lisi proposed embedding the Standard Model directly into the E_8 algebra [5]. However, the Distler-Garibaldi theorem [6] established a rigorous no-go result: any signature-preserving algebraic embedding of the Standard Model into E_8 cannot yield chiral fermions without also producing unobserved mirror partners. This theorem remains the principal obstruction to E_8 *algebraic* unification.

Here we do not employ E_8 as a gauge algebra (the effective symmetry of a quantum field theory). We instead treat the E_8 lattice as the **Geometric Substrate**, the fundamental discrete hardware on which effective dynamics emerge.

This static geometric projection dynamically assigns the Lorentzian signature $(1, 3)$ to spacetime while the internal symmetry sector retains the original Euclidean signature $(4, 0)$. This alone is sufficient to evade the no-go theorem. Mirror states therefore reside in an internal space with no timelike generator and do not propagate on the observable manifold. As detailed in appendix A, chirality emerges strictly from this geometric projection as a kinematic consequence, not as an algebraic input requiring mirror cancellation.

C. Distinction From Physics-First Approaches

Several research programs explore discrete spacetime structures (Causal Set Theory [7], Loop Quantum Gravity [8]) or emergent gravity (Verlinde [3], Jacobson [9]). While many research approaches treat spacetime or gravity as fundamental starting points, this work explores a *systems-first* methodology (deriving physical structure from universal persistence requirements rather than postulating dynamical laws). We derive universal persistence requirements from Informational Energetics and evaluate candidate substrates. We demonstrate that the E_8 lattice emerges as the optimal geometric candidate rather than an *a priori* assumption. This yields predictions that physics-first approaches do not address, while simultaneously establishing a structural rationale: the Standard Model's specific configuration (gauge groups, generation count, chirality, etc.) emerges as a consequence of persistence requirements rather than accepted as phenomenological input.

II. INFORMATIONAL ENERGETICS

Any system that persists, from a biological cell to the universe itself, must solve the same fundamental problem: how to maintain structural coherence against the constant pressure of environmental entropy. IE is a theoretical framework designed to derive the universal architecture required to solve this problem. It unifies insights from non-equilibrium thermodynamics, algorithmic information theory, and control theory into a single, predictive model of persistence.

A. The Selection Principle

Configurations populate state space transiently; not all persist. For any given level of structural complexity, those configurations that minimize their Entropic Action (S_Φ) remain observable at timescales longer than those that don't. Configurations do not *seek* minimization. The selection principle is an eliminative filter; what fails to minimize S_Φ leaves no detectable signal, and what remains constitutes the observable universe. We formalize this as:

> The **Selection Principle**: Among all configurations capable of encoding a given structural complexity, only those possessing the minimal possible **Entropic Action** (S_Φ) remain observable.

The **Entropic Action** S_Φ quantifies the total thermodynamic cost for a system to maintain structural coherence against environmental entropy over its persistence lifetime. Formally, it is the time-integrated Net Entropic Impedance:

$$S_\Phi \equiv \int_0^\tau Z_{IE}(t)\, dt \tag{1}$$

where τ is the system's lifetime and $Z_{IE}(t)$ is the instantaneous Net Entropic Impedance, the time-dependent generalization of the static pillar sum (equation (2)). For spatially distributed systems, $Z_{IE}(t)$ integrates the local dissipation density. In **Quiescent Equilibrium**, Z_{IE} is constant and $S_\Phi = Z_{IE} \cdot \tau$ is a strictly dimensionless count of elementary state transitions. **Note:** The full dynamic variational principle is deferred to the companion paper, *Informational Energetics: Entropic Action*.

Open systems (e.g., biological organisms) offset dissipation through external energy flux; closed systems minimize entropic action directly, as there is no external source to offset it.

B. The Universal Architecture: Derivation from First Principles

To persist, any system must implement a specific architecture comprising four pillars for information management and two pillars for the thermodynamic overhead of operating on its substrate.

To derive this architecture, we model a persistent entity as a **dynamic control system** that must regulate its internal state against external fluctuations (entropy).

1. The Necessary Components of Control

From control theory, any stable feedback loop requires specific functional components [10]: the Plant (the system to be controlled), the Sensor (measuring the output), the Controller (computing the correction), and the Actuator (implementing the correction), with the desired behavior specified by a Reference Signal (Setpoint).

For *robust* operation under uncertainty, control systems must additionally implement gain and phase margins to prevent instability from perturbations [11]. These margins are typically treated as tunable parameters rather than structural requirements.

Implicit in physical implementations is the irreducible energetic and temporal cost of state transitions (Landauer's Principle [1]; Margolus-Levitin limit [12]). This "Toll" is typically treated as a constraint rather than a required component.

2. The Universal Architecture of Persistence

IE reframes the control elements into six **existential pillars**, the complete, minimal set required for persistence, recontextualizing control components as survival requirements within a predictive, quantitative model. Performance comes secondary to persistence; thermodynamic and information-theoretic imperatives are elevated, necessitating a new primary unit of measure: the **Entropic Burden**. The pillars represent the synthesis of isolated foundational theorems discovered across independent disciplines and the removal of any pillar leads to catastrophic system failure (section II C).

1. **The Capacity (CAP):** *The Vessel.* The physical infrastructure required to acquire resources, process energy, and perform work. It defines the system's kinetic potential and its ability to act on the environment. (The Plant/Actuator). *Foundational Theorem (The Bekenstein Bound [13]):* Bekenstein established that a finite region of space can contain at most a finite amount of information, providing the physical basis for the strict **Finiteness** requirement of the Capacity pillar. A persistent system cannot rely on infinite state-space. Furthermore, the Margolus-Levitin limit [12] establishes that the rate at which this capacity can process state transitions is strictly bounded by its available energy.

2. **The Map (MAP):** *The Model.* The internal logic, topological structure, or compressed representation of environmental regularities that distinguishes the system from the environment. It functions as a predictive engine that reduces uncertainty (Surprisal), allowing the system to steer towards high-utility states. (The Reference Signal/Setpoint). *Foundational Theorem (The Good-Regulator Theorem [14]):* Conant and Ashby proved mathematically that "every good regulator of a system must be a model of that system." A system cannot persistently regulate against environmental entropy without encoding a structural map of its environment.

3. **The Protocol (PRO):** *The Interface.* The communicative glue regulating information flow between the Capacity and the Map. It ensures internal coherence, standardizes interfaces, and minimizes signal decay across the system's internal boundaries. (The Feedback Network/Sensor Path). *Foundational Theorem (Shannon's Source-Channel*

Separation Theorem [15]): Shannon established that optimal communication over a noisy channel requires decoupling the source data (the Map) from the error-correcting transmission encoding (the Protocol). Systemic coherence demands an independent protocol layer.

4. **The Governor (GOV):** *The Constraint.* The non-linear constraint mechanism that prevents unbounded divergence or 'runaway' loops. It enforces operational boundaries (e.g., apoptosis, budget limits) to protect the substrate from exhaustion. (Controller/Limiter). *Foundational Theorem (Ashby's Law of Requisite Variety [16]):* "Only variety can destroy variety." For a system to remain stable, its active constraint mechanism (Governor) must possess a number of states equal to or greater than the number of perturbations threatening the system.

5. **The Toll (TOL):** *The Temporal Cost.* The irreducible energy cost of state transitions and information erasure. (Processing Latency). *Foundational Theorem (Landauer's Principle [1]):* Any irreversible logical operation, such as erasing a bit, must dissipate at least $k_B T \ln 2$ of energy to the environment. State transitions carry an inescapable thermodynamic toll.

6. **The Margin (MAR):** *The Buffer.* The structural and energetic reserves held to absorb stochastic shocks. It defines the system's 'Safe Operating Area,' providing the buffer required to survive environmental variance without terminal failure. (The gain and phase margin). *Foundational Theorem (Nyquist-Bode Stability Margins [17]):* A feedback system operating at theoretical criticality has zero tolerance for variance. Nyquist proved that physical control loops must maintain strict gain and phase margins away from the critical threshold, or the first stochastic environmental fluctuation will cause terminal divergence.

Every system operates upon a **Substrate**: the physical medium imposing the immutable limits (bandwidth, latency, noise floor) within which the six pillars must function.

IE reframes control systems not as devices that execute instructions, but as structures that persist. An open-loop system merely "does": it follows a predetermined path regardless of environment. A closed-loop system "remains": it possesses the structural capacity to maintain coherence against entropic pressure. This persistence is not agency in any teleological sense; it is a **structural filter**. In a high-entropy environment, only configurations implementing the six-pillar architecture survive observation; all others dissipate.

The mapping of IE to specific physical domains is inherently adaptive. While the present work demonstrates this on the substrate of the vacuum, the IE framework states that all systems that persist are specific solutions to this same universal architecture.

3. The Entropic Balance Sheet

To quantify persistence, we distinguish between dissipation and coordination: structural burdens increase the system's entropic footprint (positive contributions to Z_{IE}), while organizational mechanisms reduce net dissipation relative to the uncoordinated baseline (negative contributions). This parallels the negative feedback signal in control theory, where the error $\epsilon = R - F$ is reduced by the feedback term.

- **Structural Costs (+):** The energetic overhead of existence. This includes maintaining boundaries (Z_{CAP}, Z_{MAR}), storing internal models (Z_{MAP}), and the irreversible cost of state updates (Z_{TOL}). These terms increase impedance.

- **Organizational Gains (−):** The reduction in dissipation achieved through coordination. The Protocol (Z_{PRO}) minimizes friction by standardizing interfaces; in control theory, this is analogous to the negative feedback signal subtracted from the setpoint ($\epsilon = R - F$). The Governor (Z_{GOV}) prevents wasteful divergence. These represent structural optimizations: the measurable difference in dissipation between a disorganized system and an organized one.

In Quiescent Equilibrium, the six pillars operate on orthogonal, statistically independent domains. By the principles of algorithmic information theory, independent channels combine additively: non-linear cross-terms represent mutual information, which vanishes when channels are orthogonal. This strictly diagonalizes the lowest-order action. The **Net Entropic Impedance** (Z_{IE}) is therefore the algebraic sum of these contributions:

$$\begin{aligned} Z_{IE} = &\underbrace{(Z_{CAP} + Z_{MAP} + Z_{TOL} + Z_{MAR})}_{\text{Structural Costs}} \\ &- \underbrace{(Z_{PRO} + Z_{GOV})}_{\text{Organizational Gains}} \end{aligned} \tag{2}$$

C. Necessity Argument: The Failure Modes

The necessity of the set of pillars $\mathbb{P} = \{\text{CAP, MAP, PRO, GOV, TOL, MAR}\}$ follows by analyzing the **Counterfactual Failure Mode** of a system whose architecture lacks exactly one pillar ($\mathbb{P}' = \mathbb{P} \setminus \{x\}$). In every case, the expected lifetime of the system τ tends to zero.

- **No Capacity:** *Starvation.* Detects threats but lacks energy to respond. ($\tau \to 0$ via Equilibrium).

- **No Map:** *Dissolution.* Lacks self-definition; random capacity activation maximizes internal entropy. ($\tau \to 0$ via Boundary Loss).

- **No Protocol:** *Decoherence.* Components decouple; actions are based on statistically independent, outdated states. ($\tau \to 0$ via Fragmentation).

- **No Governor:** *Divergence.* Unconstrained positive feedback loops exceeds structural limits. ($\tau \to 0$ via Explosion).

- **No Toll:** *Zeno Collapse.* State transitions require irreducible time and energy; the Margolus-Levitin limit establishes a strict minimum time per orthogonal state transition[12]. Without this temporal cost, updates occur instantaneously, requiring infinite bandwidth and collapsing the discrete causal sequence. ($\tau \to 0$ via Causal Collapse).

- **No Margin:** *Fragility.* Operating at theoretical criticality leaves zero buffer for environmental variance; the first fluctuation causes failure. ($\tau \to 0$ via Random Shock).

Since the removal of any component results in termination, the set $\mathbb{P}$ is **Minimally Necessary**.

D. Scope of the Architecture

The six pillars map onto the canonical components of a control loop, augmented by constraints from information theory and thermodynamics that standard control theory treats as external parameters. Advanced topologies (feedforward networks, state observers) extend this basic structure but do not introduce new primitive requirements for persistence.

We conjecture that the six pillars are sufficient for the systems analyzed here. Complex functions often cited as distinct requirements such as Memory (persistence of MAP over time, MAR) or Prediction (MAP operation via PRO) are treated as emergent properties of the fundamental set. Whether this set is mathematically minimal across all possible persistent systems remains an open question; the framework's predictive power provides empirical support for its sufficiency in the domain of fundamental physics.

E. Domain of Application and Boundary Limits

Complex Adaptive Systems (CAS) research has historically yielded rich qualitative taxonomies. IE introduces a falsifiable paradigm shift: by applying the constraints of robust control theory to CAS, and strictly isolating systems that successfully **persist** against entropy, open-ended qualitative observation is replaced by exact structural requirements.

While macroscopic domains (biological, ecological, economic) must manage these six existential pillars probabilistically within chaotic, open environments, this work deliberately operationalizes the framework at the absolute boundary limit of fundamental physics. The vacuum serves as the ultimate zero-external-flux baseline, a sterile analytical control environment required to prove the architecture's exact mathematical validity before evaluating the time-varying, statistical distributions of capacity and margin found in open macroscopic systems.

III. SYSTEM 0: ARCHITECTURAL REQUIREMENTS FOR PERSISTENCE: FINITENESS, UNITARITY, CAUSALITY

If the principles of IE are truly universal they must apply to the most fundamental layer of reality, the vacuum itself. In this section we will map the six pillars to this specific domain resulting in three non-negotiable requirements for the substrate of reality:

- **Finiteness:** $(\mathrm{CAP}, \mathrm{MAR})$ To prevent energetic and informational divergence.

- **Unitarity:** $(\mathrm{MAP}, \mathrm{PRO})$ To ensure the lossless conservation of information.

- **Causality:** $(\mathrm{GOV}, \mathrm{TOL})$ To enforce a well-defined temporal evolution.

A. The Closure Constraint

Within IE the vacuum is the minimal persistent substrate from which all observable structures emerge. Because no deeper persistent system exists beneath the vacuum, it cannot obtain configuration information, boundary conditions, synchronization signals, or memory from an external source and all information required to specify its state must be internally generated. We refer to this requirement as the **Closure Constraint**: the substrate of reality must be entirely self-defining. Its geometry must arise from its own architectural requirements. *Any candidate substrate requiring outside data to specify its initial conditions is not a fundamental ground state.*

B. Persistence Requires $\mathrm{CAP}, \mathrm{MAR}$ via Finiteness

By the Closure Constraint the vacuum must be self-specifying. A self-specifying system cannot require infinite information to describe any local neighborhood as that would need an external decompressor that does not exist. Even with infinite storage, evaluating an infinitely complex model would take infinite time: each bit of the minimally compressed representation must be consulted to guarantee Governor containment, and any finite processing rate implies divergent latency. The system would be paralyzed before the next fluctuation arrives. Finiteness is therefore mandatory. The Capacity pillar enforces a finite state-space; the Margin pillar enforces finite reserves against fluctuation. Together we now show that finiteness require the substrate to be discrete, homogeneous, and a lattice.

1. Discrete (Indivisible Unit)

A continuous volume admits infinite descriptive complexity. Under the Closure Constraint, no external algorithm exists to compress this to finite Kolmogorov complexity, so a continuous substrate cannot be self-defining. Therefore, the substrate must be **Discrete**: there must exist a fundamental, indivisible unit of information that sets a maximum density.

2. Homogeneous (Direction-Independent)

To minimize algorithmic complexity, the substrate must be **Homogeneous**. In Algorithmic Information Theory, the structural complexity of a random graph scales with its size ($K(S) \propto N$). For the universe to be scalable without additional processing overhead, its Kolmogorov Complexity must be $O(1)$, independent of size (Closure Constraint). This requires that the local topology be the same regardless of location and direction.

3. Lattice (Discrete + Homogeneous)

A structure that is both discrete (finite information density) and homogeneous (constant structure in every direction) is a **Lattice**. This excludes continuous manifolds (infinite description) and random graphs (non-constant complexity).

C. Persistence Requires MAP, PRO via Unitarity

For the vacuum to persist, it must maintain a stable Map over time. This requires that information is perfectly conserved. By the Closure Constraint (section III A), the vacuum must be a perfectly closed and lossless information network, making Unitarity a structural necessity, corresponding to the Map and Protocol pillars. This requirement imposes four structural properties on the substrate:

1. Lattice Must Be Positive Definite

Information-Theoretic Justification: In a metric space, the norm $|\mathbf{v}|^2 = g_{\mu\nu}v^\mu v^\nu$ represents the information distance from the origin (reference state). For the system to have a well-defined minimum energy configuration (stable ground state), this distance must satisfy:

$$|\mathbf{v}|^2 \geq 0 \quad \forall \mathbf{v} \neq 0 \tag{3}$$

A metric with negative eigenvalues (Lorentzian) allows $|\mathbf{v}|^2 < 0$, implying an imaginary "distance" from the reference state and preventing the definition of a stable ground state. Furthermore, a Lorentzian metric admits non-trivial null vectors ($|\mathbf{v}|^2 = 0$ where $\mathbf{v} \neq 0$), allowing for the creation of infinite information density without exceeding the capacity budget ($|k\mathbf{v}|^2 = 0$ for any scalar amplitude k), which violates the **Finiteness** pillar. Therefore, the **substrate** must be Euclidean.

The Substrate-Projection Dual: The Euclidean positive-definiteness applies strictly to the substrate lattice. The observed Lorentzian signature $(-,+,+,+)$ emerges dynamically from the causal projection derived later in section IV D; the substrate memory serves as a static, definite metric, while the active processor operates via an indefinite metric. This fundamental distinction provides the exact geometric mechanism by which time acquires its directional arrow, satisfying a core causal requirement of the persistence architecture.

2. Lattice Must Be Mappable to a Torus

To satisfy **Finiteness**, the system must be spatially bounded. To satisfy **Protocol** (Unitarity), it must be closed (no edges). By the combinatorial Gauss-Bonnet theorem, a regular lattice of fixed vertex coordination tiled on a closed surface of Euler characteristic χ requires topological defects if $\chi \neq 0$. The sphere ($\chi = 2$) therefore requires defects that break translational invariance. The Euclidean **Torus** (T^n, $\chi = 0$) provides the unique flat, closed topology with periodic boundary conditions that admits a homogeneous, defect-free lattice structure, simultaneously satisfying Finiteness (bounded), Unitarity (closed), and Homogeneity (flat).

Consequently, the statistical evolution of the vacuum is defined by the Partition Function on a torus[1]:

$$Z(\beta) = \sum_{\text{states}} e^{-S_{\text{config}}} = \mathrm{Tr}(e^{-\beta H}) \tag{4}$$

3. Lattice Must Be Even

A stable Map requires path independence, which in turn requires evenness. A persistent system demands a unique, unambiguous equilibrium state. If the state

[1] Here, β is the formal periodicity parameter of the imaginary time cycle. We use the partition function formalism for state enumeration on a torus, not to claim the vacuum has a literal temperature.

count Z depended on the path taken through moduli space (parameterization) rather than the configuration itself, the vacuum would lack a stable **Map**.

This requires $Z(\beta)$ to be single-valued under the modular transformation $T : z \to z + 1$. The Jacobi Theta Function, $\Theta_\Lambda(z) = \sum e^{i\pi z|\mathbf{v}|^2}$, counts these states. For **Even** lattices ($|\mathbf{v}|^2 = 2n$), the exponent $2\pi inz$ is invariant under the shift. However, any odd-norm vector ($|\mathbf{v}|^2 = 2n + 1$) introduces a sign inversion ($\Theta \to -\Theta$), rendering the Map multi-valued. Thus, the lattice must be **Even**.

4. Lattice Must Be Self-Dual and Unimodular (Read/Write Symmetry)

To prevent information loss via destructive interference or Landauer erasure, the encoding operation (write) and the decoding operation (read) must be informationally equivalent. This is enforced by the modular transformation $\mathcal{S} : z \to -1/z$, which maps the lattice to its reciprocal (Fourier dual).

By the Poisson Summation formula, the partition function transforms as:

$$\Theta_\Lambda(-1/z) \propto \frac{1}{\mathrm{vol}(\Lambda)} \Theta_{\Lambda^*}(z) \tag{5}$$

For the Map to remain invariant ($Z_\Lambda = Z_{\Lambda^*}$), the lattice must be **Self-Dual** ($\Lambda = \Lambda^*$). This strictly enforces **Unimodularity** ($\mathrm{vol}(\Lambda) = 1$), as the only scalar satisfying $V = 1/V$ is 1. Any other volume introduces an irreversible scaling factor, violating Unitarity.

Requirements: The lattice of the vacuum must be **Positive Definite, Unimodular, Even, and Self-Dual**.

D. Persistence Requires $\mathrm{GOV}, \mathrm{TOL}$ via Causality

For the vacuum to persist, it must exist stably and *evolve* in a well-defined manner. This creates a fundamental information-theoretic challenge: how to project the vast state space of a lattice node onto a single, linear temporal axis without ambiguity.

The Serialization Problem: From the **Capacity** pillar, we establish that any persistent system must possess a finite state-space dimension, denoted N (the number of linearly independent, orthogonal basis vectors spanning the node's total informational payload, i.e., the Hilbert-space dimension). To evolve, the system must serialize these N potential states onto a discrete timeline defined by the **Temporal Modulus** (Δ), representing the number of available temporal slots in one fundamental causal cycle.

Causal Aliasing: To express its full informational capacity, the system must serialize all N potential states into Δ discrete temporal slots per fundamental cycle. By the Pigeonhole Principle, if $N > \Delta$, the number of states exceeds the available temporal slots. Consequently, at least one temporal coordinate must simultaneously host multiple distinct states. This results in **Causal Aliasing**: distinct information states become concurrently superimposed in time, destroying the system's ability to maintain a well-defined serial history.

The Persistence Inequality: To enforce a strict arrow of time and satisfy the Causality requirement, the projection must satisfy the Persistence Inequality: $\Delta \geq N$. The temporal container must meet or exceed the information content it holds. While the inequality allows $\Delta > N$, the Closure Constraint forbids inert auxiliary overhead. Therefore, any excess temporal capacity ($M = \Delta - N$) cannot be arbitrary; it must be explicitly mandated by additional structural requirements to maintain coherence.

E. Summary: The Architectural Requirement of the Vacuum

Through the application of Informational Energetics, we have fully specified the architectural requirements for the substrate of reality:

1. $(\mathrm{CAP}, \mathrm{MAR})$ **Finite Lattice**, required by Finiteness.

2. $(\mathrm{MAP}, \mathrm{PRO})$ **Positive Definite, Unimodular, Even, and Self-Dual**, required by Unitarity.

3. $(\mathrm{GOV}, \mathrm{TOL})$ Any causal system must satisfy a **Causal Projection**: a serialization of its N-dimensional state-space onto a discrete timeline of length $\Delta \geq N$.

With this specification complete, the physical substrate is a constrained mathematical problem: find the unique geometric object that satisfies all of these requirements simultaneously.

IV. SYSTEM 1: THE PROJECTION, THE $E_8 \to D_4 \oplus D_4$ GEOMETRIC PROJECTION THAT UNIQUELY SATISFIES THE SYSTEM 0 REQUIREMENTS

Having established the requirements of the substrate we now identify the unique mathematical structure that satisfies these constraints.

A. The Lattice Selection (E_8)

We will begin by determining the possible dimensions of the lattice, then identify a unique lattice, and finally determine its projection into spacetime.

1. The $n \equiv 0 \pmod 8$ Constraint

The requirement of an even, self-dual lattice is remarkably restrictive. A fundamental constraint from the theory of modular forms and lattices[18], states that even, self-dual lattices only exist in dimensions that are multiples of 8. This immediately eliminates the majority of dimensionalities leaving only:

$$n \in \{8, 16, 24, \dots\}$$

This eliminates any lattice solution in dimensions $n < 8$. Consequently: the observed $n = 4$ spacetime cannot mathematically host a unitary, self-dual information substrate. The universe we observe cannot be the fundamental hardware; it must be an emergent projection of a higher-dimensional structure.

Furthermore, this excludes $n = 10$ as a standalone geometric substrate. Because $10 \not\equiv 0 \pmod 8$, a 10-dimensional lattice cannot intrinsically satisfy *Unitarity* (the requirement for an even, self-dual lattice). While continuous theories can recover Unitarity via auxiliary structures (e.g., Calabi-Yau compactifications), selecting a specific manifold requires external configuration memory, which is prohibited by the strictly self-defining Closure Constraint (section III A).

2. The Principle of Algorithmic Determinism

To satisfy **Map** and **Unitarity**, the substrate geometry must be intrinsic and self-defining. It cannot depend on arbitrary selection parameters or external boundary conditions. We demonstrate that the constraints on the possible dimensions eliminate all but one unique option:

1. **Unitarity** mandates even, self-dual, unimodular lattices. By Kneser's theorem, such lattices exist only for $n \equiv 0 \pmod 8$.
2. By the **Closure Constraint** (section III A), the vacuum admits no external configuration memory. The algorithmic specification complexity of the substrate's geometry is therefore **zero bits** beyond the architectural constraints derived in System 0.
3. At $n = 8$: The E_8 lattice is the **unique** even self-dual lattice [18]. The algorithmic specification complexity of a uniquely determined geometry is $\log_2(1) = \mathbf{0}$ **bits** satisfying the Closure Constraint.
4. At $n = 16$: The architectural constraints isolate exactly two non-isomorphic solutions ($E_8 \oplus E_8$ and D_{16}^+). Isolating a unique geometry from this degenerate set requires $\log_2(2) = \mathbf{1}$ **bit** of additional specification (applied strictly to the ground-state definition, not to dynamical excitations). Because the Closure Constraint enforces **zero bits** of external specification, this substrate is not self-specifying.
5. At $n = 24$: Twenty-four distinct Niemeier lattices exist, requiring $\log_2(24) \approx 4.58$ bits of specifying information to resolve the degeneracy, similarly to violating the Closure Constraint.
6. The number of distinct even self-dual lattices grows extremely rapidly with dimension. For example, in dimension 32, there are more than 80 million such lattices.

The E_8 lattice is selected not merely for its low dimensionality, but because it is the **only self-dual geometry that requires zero bits of selection information**. It is uniquely self-defining. This differs from the Δ derivation, where surjection and continuum limit are forced by causal continuity and atomicity (physical requirements) not arbitrary choices between equivalents.

This unique self-consistent solution is selected prior to the emergence of time, eliminating any argument of spontaneous symmetry breaking which would require fluctuations, transition rates, and a dynamical framework that does not yet exist and which the substrate must first generate.

The optimality of the E_8 lattice extends beyond self-duality: Viazovska's proof establishes it as the densest sphere packing in eight dimensions [19], the unique solution to a geometric extremization problem. This reinforces its status as the substrate requiring zero selection information. E_8 is self-defining and geometrically optimal.

B. The Projection ($D = 4$ and $\nu = 16$)

The E_8 lattice cannot process information by itself. Processing requires a flow (Input vs. Output). The projection must break the E_8 symmetry to distinguish the "Observer" (Spacetime) from the "System" (Internal States).

1. The Symmetric Decomposition (External Spacetime vs. Internal Symmetry)

The projection must be compatible with the self-duality of the parent E_8 lattice to maintain local probability conservation. Because E_8 is self-dual ($\Lambda = \Lambda^*$), any symmetric decomposition into orthogonal sublattices ($E_8 \to L_4 \oplus L_4$) must admit a specific glue group whose coset structure exactly reconstructs the self-dual parent. To avoid unbalancing the Unitarity requirement, the state space must be partitioned into two symmetric sectors that can support this exact quotient structure.

Candidate Rank-4 Subsectors. We evaluate rank-4 root sublattices that can symmetrically embed in E_8. The irreducible rank-4 root systems are A_4, B_4, C_4, D_4, and F_4. We restrict our search to sublattices admitting a symmetric embedding $E_8 \to L_4 \oplus L_4$. B_4 and C_4 violate the Evenness requirement because their root lattices contain roots of two distinct lengths. F_4, while even, is not self-dual. This leaves $A_4 \oplus A_4$ and $D_4 \oplus D_4$ as the mathematically viable candidates for the symmetric decomposition of the substrate.

- **The A_4 No-Go (Closure Mismatch):** The $A_4 \oplus A_4$ orthogonal decomposition has an index of 5 in E_8, yielding a cyclic glue group $\mathbb{Z}_5$. Under the **Closure Constraint** (Section III A), the substrate admits no extraneous structural data. Because $\mathbb{Z}_5$ is cyclic of prime order, its states cannot be partitioned into a symmetric, self-contained addressing scheme; resolving the quotient structure would require an externally specified translation table, violating the requirement that the vacuum be entirely self-defining.

- **The D_4 Solution (The Unitary Partition):** The $D_4 \oplus D_4$ orthogonal decomposition has an index of 4 in E_8, yielding a glue group $(\mathbb{Z}_2 \times \mathbb{Z}_2)$ composed of involutions. This structure supports a symmetric, self-contained addressing scheme with zero external specification, satisfying the Closure Constraint. Furthermore, D_4 is the unique rank-4 even lattice that supports **Triality**, providing the exact geometric symmetry required to construct the Map and Causality pillars (section IV B 2).

Therefore, $E_8 \to D_4 \oplus D_4$ is the **strictly unique symmetric decomposition** that preserves global Unitarity while naturally generating the chiral structure required for information processing.

This unique decomposition partitions the 8 dimensions into two orthogonal sectors with distinct roles:

1. **Sector A (External Spacetime):** The first D_4 lattice defines the coordinate addresses of the lattice nodes. Since the D_4 root system has rank 4 (four linearly independent simple roots), it naturally projects onto a 4-dimensional coordinate manifold, fixing **D=4** for spacetime. The rank equals the number of independent Cartan generators, which specify the coordinate addresses; hence rank 4 implies $D = 4$.

2. **Sector B (Internal Symmetry):** The second D_4 lattice encodes the internal configuration of each node, its state within the substrate's symmetry structure. These four dimensions are internal degrees of freedom and not spatial directions.

This structural partition explains why the substrate must eventually appear 4-dimensional while possessing complex internal forces, without requiring the hidden spatial dimensions of Kaluza-Klein theory.

2. *The Bit-Depth of the Node ($\nu = 16$)*

To determine the information capacity (Bit-Depth) of the lattice nodes we ask: what is the minimal geometric structure required to define a persistent, distinguishable signal on the lattice?

Geometric Translation: In a lattice, information states are encoded as **orientations**—directional configurations that can be rotated and transformed. Because these transformations are strictly geometric rotations of a coordinate space, they are governed by the Orthogonal groups $SO(n)$ and their simply-connected double covers, the Spin groups $Spin(n)$. While other Lie groups (such as $SU(n)$) admit complex representations, they describe abstract internal symmetries; the physical orientation of the substrate is strictly bound to $Spin(n)$ geometry.

The D_4 lattice spans 4 spatial dimensions, but its associated Lie algebra ($\mathfrak{so}(8)$) acts on 8-dimensional fundamental and spinor representations. As established in section IV B, each D_4 factor in the $E_8 \to D_4 \oplus D_4$ decomposition therefore provides a local continuous rotation symmetry of $Spin(8)$. We must determine the minimal channel capacity (ν) required for this native $Spin(8)$ hardware to satisfy the **Map** and **Causality** pillars.

1. **The Map Constraint (Complex Phase):** While $Spin(8)$ possesses chiral spinors, they are mathematically **Real** (Majorana-Weyl). In signal processing terms, this means a state vector is identical to its conjugate ($\psi = \bar{\psi}$). A system built strictly on an 8-channel real representation cannot distinguish a signal from its inverse (Phase Ambiguity) because it lacks complex phase information. To support a stable Map, the system requires **Complex Representations** ($\psi \neq \bar{\psi}$). The minimal encoding unit is the hardware's native spinor dimension (8), set by the $D_4/\mathfrak{so}(8)$ algebra; no sub-8 representation supports the full rotational structure of the substrate. A complex representation of this base dimension requires 16 real degrees of freedom. Because the native $Spin(8)$ spinors are real, encoding a complex phase distinct from amplitude demands a direct sum of two independent 8-dimensional real channels, yielding $8+8 = 16$ real channels.

2. **The Causality Constraint (Chirality):** To support the **Causality** pillar (Arrow of Time), the system must distinguish "Input" from "Output." Geometrically, this requires **Chirality**, the ability to distinguish Left-handed projections from Right-handed projections, preventing information from reflecting symmetrically across the causal loop.

To satisfy both constraints simultaneously, the node utilizes the representations natively available to its D_4 ($\mathfrak{so}(8)$) geometry. Because D_4 possesses **Triality**, its geometry uniquely supports independent 8-dimensional chiral representations: the Left-handed spinor (8_L) and the Right-handed spinor (8_R).

To encode a persistent, complex, causal signal, the node must allocate exactly one chiral representation for input (8_L) and one for output (8_R). This structurally enforces Causality while jointly providing exactly the 16 real channels required to encode complex phase (Map).

The total internal channel capacity (Bit-Depth) of the node is therefore the direct sum of these paired representations:

$$\nu = \dim(8_L) + \dim(8_R) = 8 + 8 = \mathbf{16} \tag{6}$$

By deriving $\nu = 16$ strictly from the rank-4 D_4 hardware, we satisfy the channel capacity requirements without requiring embedding in a higher-dimensional gauge group.

C. The Total Node Capacity (N)

We derive the total information capacity of a single lattice node, which constrains the state space of all subsequent derivations.

The projection $E_8 \to D_4 \oplus D_4$ creates two orthogonal, symmetric subsystems. As derived above, the internal sector requires a bit-depth of $\nu = 16$. Because the projection must be compatible with the **Self-Duality** of the parent E_8 substrate, the Spacetime sector must remain informationally symmetric to the Internal sector. Because the parent E_8 substrate is exactly self-dual ($\Lambda = \Lambda^*$), the projection cannot introduce an informational asymmetry between the spacetime manifold and the internal state space without violating local probability conservation. Therefore, the maximal channel capacity supported by the geometric structure of Sector B strictly dictates the required reciprocal capacity of Sector A. Because information capacity (bit-depth) is additive across independent orthogonal sub-spaces, the total channel capacity N is simply the sum of the two sectors:

$$N = \nu_{\text{Sector A}} + \nu_{\text{Sector B}} = 16 + 16 = \mathbf{32} \tag{7}$$

This integer $N = 32$ represents the total number of distinct state channels available for encoding information on the lattice substrate.

D. Dimensional Reduction: Geometric Origin of The Metric Signature and Causality

1. The Requirements

To map the 16-channel signal onto a 4-dimensional manifold without loss or ambiguity, the metric algebra must satisfy two strict encoding constraints:

1. **Complex Compression (Map):** The 16 real channels must be compressed into 8 complex degrees of freedom ($16\mathbb{R} \to 8\mathbb{C}$). This requires the algebra to naturally generate a geometric imaginary unit (i) to encode phase information.

2. **Chiral Sorting (Causality):** The system must distinguish "Input" from "Output" to enforce the arrow of time. This requires the algebra to support orthogonal projectors (P_L, P_R) that split the signal into directed halves ($8\mathbb{C} \to 4\mathbb{C}_L + 4\mathbb{C}_R$).

2. The Search Space

A 4-dimensional manifold admits two metric signatures. We test each against the requirements:

- **Option A: Euclidean Metric** $(+,+,+,+)$**.** In a space where all dimensions are spatial, the volume element (pseudoscalar ω) squares to positive unity ($\omega^2 = +1$).

- **Option B: Lorentzian Metric** $(-,+,+,+)$**.** In a space with one temporal dimension, the Clifford algebra volume element (the pseudoscalar ω) squares to negative unity ($\omega^2 = -1$).

3. The Unique Solution

The Euclidean metric cannot satisfy the dynamic encoding constraint. Because $\omega^2 = +1$ in signature $(+,+,+,+)$, the center of the Clifford algebra is strictly real ($\mathbb{R} \oplus \mathbb{R}$). It lacks the capacity to generate the global geometric imaginary unit i required for complex phase compression, nor can it support complex chiral projectors. A universe with this signature would be a static, real-valued block with no capacity for phase or flow.

The **Lorentzian Metric** is the unique valid solution. While the $(2,2)$ split signature yields $\omega^2 = +1$ (restoring a real algebra), the $(1,3)$ signature yields $\omega^2 = -1$. Because the Clifford volume element squares to negative unity, it functions identically as the global geometric imaginary unit ($i \equiv \omega$). This naturally generates the complex structure required to compress the signal ($\nu = 16$) and the chiral structure required to sort it.

Crucially, the metric projection assigns the Lorentzian signature $(-,+,+,+)$ exclusively to Sector A (Spacetime). Sector B (Internal Symmetry) retains the original Euclidean signature $(+,+,+,+)$ of the E_8 algebra. This asymmetry is the direct consequence of the Causality requirement: only one D_4 factor can support a timelike generator, and that assignment uniquely distinguishes the spacetime sector from the internal sector.

Conclusion: The signature $(-,+,+,+)$ is the only algebraic structure capable of processing the system's information content and is not an arbitrary choice. Physically, the negative sign of Time ($dt^2 < 0$) is the geometric cost required to generate the imaginary unit i.

a. Physical Consequence: Once these algebraic structures are established to satisfy the encoding requirement, they manifest in physics as fundamental properties of matter. The **Matter/Antimatter** distinction emerges from the complex conjugation ($\psi \leftrightarrow \bar{\psi}$) enabled by the imaginary unit. **Chirality** emerges from the orthogonal projectors. These are unavoidable consequences of encoding a 16-channels onto a 4-dimensional Lorentzian manifold.

E. The System Logic (σ=5 and χ=2)

To satisfy the Map pillar, a persistent entity must strictly distinguish its internal state from the ambient environment. However, the **Finiteness** and **Unitarity** pillars require the vacuum to be homogeneous (no preferred positions) and self-dual (informationally symmetric). For a localized state to possess both a coordinate address and a distinguishable Map while satisfying these background constraints, the two specifications must be orthogonal: independent, yet simultaneously well-defined. Geometrically, this requires the substrate to support a topological defect, a localized boundary structure capable of separating a region from the vacuum without violating the background homogeneity. The minimal embedding dimension for this orthogonal decomposition sets the rank of the interaction symmetry (σ), the geometric degrees of freedom available for force mediation.

1. The Topological Boundary ($\chi = 2$)

To satisfy the **Binary Partition Constraint**, a persistent entity must be an **extended closed boundary structure**: a circulation of information confined to a finite region, with a boundary that rigorously separates "Inside" from "Outside". The minimal such boundary is the unique simply-connected closed surface, the sphere (S^2, $\chi = 2$, $g = 0$). A defect is therefore not a point-like flaw.

We derive this by analyzing the Euler Characteristic for closed, orientable surfaces:

$$\chi = 2 - 2g \tag{8}$$

where g is the genus (number of holes).

- **The Exclusion of $g \geq 1$ (The Leaky Partition):** Any topology with one or more holes (Torus $g = 1$, Double Torus $g = 2$, etc.) fails to define a strict binary separation.
  - *Information-Theoretic Failure:* A genus $g \geq 1$ surface is not simply connected. It supports non-contractible loops, closed paths that thread through the holes without intersecting the surface. This creates informational ambiguity: information flux can traverse the topology without being strictly partitioned into "Inside" or "Outside," rendering the total conserved quantity associated with the boundary ill-defined.
  - *Thermodynamic Failure:* By the Gauss-Bonnet theorem ($\int_M K\, dA = 2\pi\chi$), any surface with $g \geq 1$ has $\chi \leq 0$, implying neutral or negative total curvature. Such a surface cannot support net positive entropic pressure (information density) against the bulk substrate: it would structurally collapse.
- **The Uniqueness of $g = 0$ (The Sphere):** The sphere is the unique closed surface with $g = 0$, yielding $\chi = 2$, the maximum possible value. It is the only simply connected topology, ensuring that all loops contract to a point. This forces all information flux to be either contained or excluded, enabling the perfect binary partition of state required for a persistent, distinguishable topological defect.

a. Forward Implication (System 2 Consequence): The invariant $\chi = 2$ is the *necessary and sufficient* condition for **charge quantization**. Because χ must be an integer and $\chi = 2$ is the unique maximum for closed surfaces, the charge associated with this topology is discrete. You can have 1 sphere or 2 spheres, but not 1.5 spheres. This geometric constraint allows the continuous lattice field to support discrete, countable units of charge.

2. The Embedding Invariant ($\sigma = 5$)

We derive σ as the minimal coordinate set required to fully specify a persistent entity on the lattice. A localized state requires two independent pieces of information: its **spatial address** (where it is) and its **topological boundary** (what it is).

The Orthogonality Condition: A fundamental requirement of the Homogeneity defined in System 0 is that the internal definition of a topological defect (its boundary topology) must be entirely independent of its location. Mathematically, this requires the total configuration space to be a Cartesian product of the spatial manifold and the internal boundary manifold ($V_{total} = V_{space} \times V_{boundary}$). By definition, the dimension of a product manifold is the sum of the dimensions of its factors. This additive combination is a geometric necessity, not an externally selected architectural choice.

The dimensions of these two orthogonal factors are uniquely constrained:

- **Spatial Freedom ($D_{space} = 3$):** The **Causality** pillar requires that one dimension of the $D = 4$ manifold be allocated to the update stream (Time). This leaves exactly $D - 1 = 3$ dimensions for spatial configuration. This value is also structurally mandated by the boundary topology: the persistent defect possesses an S^2 boundary ($\chi = 2$), which cannot be geometrically embedded in a spatial manifold of fewer than 3 dimensions.

- **Topological Boundary ($D_{boundary} = 2$):** The **Map** pillar requires the topological defect to be distinguished from the vacuum by a closed boundary. As derived in section IV E 1, the unique simply-connected solution is the sphere (S^2). Regardless of its location in the 3D spatial bulk, specifying the internal state of this boundary requires exactly 2 intrinsic coordinate parameters (e.g., θ, ϕ).

Because these two parameter spaces are strictly orthogonal, the dimension of the minimal geometric embedding is the exact sum of these independent sectors:

$$\sigma = \dim(V_{space}) + \dim(V_{boundary}) = 3 + 2 = \mathbf{5} \qquad (9)$$

Note: $\sigma = 5$ is the minimal *geometric embedding* dimension for a persistent defect (spatial address plus boundary coordinates), not the gauge-group rank of an effective quantum field theory. The mapping from this geometric embedding to observed force quantum numbers is derived dynamically in the companion paper on Entropic Action.

F. The Fundamental Resonance ($\Delta = 43$)

To derive the fundamental clocking period (Δ) of the lattice's causal cycle, the system must simultaneously satisfy three independent physical constraints. These constraints uniquely isolate the substrate's resonant cycle:

- Unitarity restricts Δ to the Stark-Heegner numbers.
- Causality requires $\Delta \geq N = 32$.
- Self-Clocking requires $\Delta \leq 2N = 64$.

1. *Unitarity and The Unique Factorization Domain Constraint*

For a system to preserve quantum Unitarity its causal history must be uniquely reconstructable; multiple distinct causal histories cannot blend into identical outcomes, or information is irreversibly lost.

Mathematically, a $(1+1)D$ causal sequence on a self-dual lattice defines an elliptic curve. As proven formally in appendix B, for the evolution operator of this curve to be uniquely invertible (history is perfectly preserved), the curve's endomorphism ring must be a Principal Ideal Domain. This requires the algebraic structure to possess *Complex Multiplication (CM) with a class number of $h =$ 1.*

To preserve the self-duality of the substrate, the geometric period of the loop (Δ) is identified with the algebraic CM discriminant (D). Consequently, the Unitarity requirement restricts the clocking period Δ to the Stark-Heegner numbers (the absolute values of the fundamental discriminants of the nine imaginary quadratic fields of class number one): $\Delta \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$.

2. *Causality (The "Bandwidth") Constraint*

To satisfy Causality we apply the constraint $\Delta \geq N = 32$ (established in section III D) leaving: $\Delta \in \{43, 67, 163\}$

3. *The Self-Clocking Constraint (Causal Continuity)*

By the Closure Constraint, the vacuum has no external reference clock; time must advance relationally through state transitions. In a periodic sequence of Δ steps with $N = 32$ active signal states and $M = \Delta - N$ empty overhead slots, a run of k consecutive empty slots creates a k-fold ambiguity in elapsed time. The node cannot distinguish "waiting 1 step" from "waiting k steps."

To resolve this ambiguity, the system cannot reference an external clock (Closure), cannot allocate channel capacity to an internal memory counter (Finiteness), and cannot rely on spatial neighbors as a surrogate clock because translational Homogeneity renders all local neighbors geometrically identical during static intervals. To maintain temporal synchronization, the maximum run of empty slots must be strictly bounded to 1.

This is achievable if and only if $M \leq N$: When $M > N$, the Pigeonhole Principle guarantees that no arrangement of M empty slots on a circular sequence of length Δ can avoid adjacent pairs; consecutive empty slots are structurally forced. Therefore, this run-length limited constraint[20] (which forbids arbitrary sequences of identical symbols) strictly requires $N \geq M$, yielding:

$$N \geq (\Delta - N) \implies \Delta \leq 2N = 64 \qquad (10)$$

Result: With $N = 32$, the upper limit for a self-clocking fundamental cycle is $\Delta \leq 64$. This eliminates the remaining Stark-Heegner candidates 67 and 163, leaving $\Delta = 43$ as the strictly unique geometric solution for the temporal period.

G. Manifold Resolution

Given that the substrate is a $D = 4$ dimensional manifold with fundamental temporal cycle $\Delta = 43$, the theoretical maximum number of distinct spacetime slots the local geometry can support per fundamental update cycle is the product:

$$R_M = D\Delta = 4 \cdot 43 = 172 \quad (11)$$

Persistent topological defects occupy subsets of this address space; their operational costs in System 2 scale with the fraction of R_M utilized.

This completes the substrate's information geometry: $\nu = 16$ encodes what a single spinor holds, $N = 32$ the total channels at a node, and $R_M = 172$ the spacetime addresses available per cycle.

H. The Entropic Loads: Resolution Limits of the Vacuum

The vacuum's persistence depends on three hierarchical scales of informational complexity. Each scale defines a resolution limit, a threshold below which signals dissolve into the substrate.

1. The Intrinsic Load ($L_{intrinsic} = 23$): Internal Specification

The information required to specify a topological defect's internal state comprises three orthogonal components. The chiral configuration ($\nu = 16$) encodes phase and handedness. The interaction embedding ($\sigma = 5$) encodes the coordinate degrees of freedom: spatial address ($D_{space} = 3$) and boundary coordinates ($D_{boundary} = 2$, the continuous parameters required to specify a point on the boundary surface). The topological invariant ($\chi = 2$) encodes the global identity of the boundary, the discrete classification that distinguishes the sphere ($\chi = 2$) from all other closed surfaces, independent of local coordinates. Together these three orthogonal components set the internal resolution limit: the precision with which the vacuum can distinguish *what* a defect is (internal state) from *where* it is (spatial position).

$$L_{intrinsic} = \nu + \sigma + \chi = 16 + 5 + 2 = \mathbf{23} \quad (12)$$

2. The Embedding Load ($L_{embed} = 31$): Manifold Address

The total information to specify a defect's complete configuration: internal state plus spatial address.

Because the chiral signal states are dynamic entities within the $D = 4$ manifold, specifying their complete state address requires more than just spatial coordinates. In discrete control theory and state-space representations, uniquely specifying a dynamic state's full trajectory requires establishing both its position vector (D dimensions) and its transition rate or orientation vector (D dimensions). This necessitates a total embedding cost of $2D = 8$ independent degrees of freedom to uniquely specify the dynamic state address on the manifold.

$$L_{embed} = L_{intrinsic} + 2D = 23 + 8 = \mathbf{31} \quad (13)$$

3. The Substrate Load ($L_{substrate} = 188$)

The vacuum's intrinsic structural complexity. This sets the substrate capacity, the total information the vacuum must process per fundamental cycle.

- **$\nu = 16$:** Channel infrastructure. The spinor information at each site.
- **$R_M(D \cdot \Delta) = 172$:** Manifold Resolution.

$$L_{substrate} = (R_M) + \nu = (4 \cdot 43) + 16 = \mathbf{188} \quad (14)$$

I. The E_8-Persistence System Specification

We have now identified a unique solution that satisfies the architectural requirements of System 0.

Substrate: The E_8 Manifold. The $D_4 \oplus D_4$ decomposition of E_8 fixes $D = 4$ as the unique dimensionality supporting Finiteness and Causality.

- **Capacity: The Operational Envelope ($N = 32$).** The $N = 32$ chiral channels within each $\Delta = 43$ cycle define the total state-transition budget. The clock period sets the envelope; the channel count determines what fills it.
- **Map: The Persistent Embedding ($\sigma = 5$).** The sum of spatial ($D_{\text{space}} = 3$) and topological ($D_{\text{boundary}} = 2$) degrees of freedom required to uniquely embed a persistent entity in the manifold.
- **Protocol: The Chiral Bit-Depth ($\nu = 16$).** The minimal spinor dimension admitting complex structure and chirality, enabling lossless information propagation. Required by Unitarity for read/write symmetry.
- **Governor: The Topological Boundary ($\chi = 2$).** The sphere (S^2), unique simply-connected closed surface, enforcing binary partition of state and preventing field divergence. Required by Finiteness.
- **Toll: The causal cost of persistence ($\tau = -1$, $M = 11$).** Structurally, the Lorentzian metric ($\tau = -1$) encodes time by sacrificing one spatial dimension's definiteness. Operationally, the $M = \Delta - N = 11$ overhead intervals in the fundamental cycle provide the temporal slack required for lossless state transitions.

- **Margin: The Resolution Floor ($L_{embed} = 31$).** The total embedding dimension sets the geometric noise floor below which signals dissolve into the substrate. In Quiescent Equilibrium, this floor is **saturated**: the minimal resolvable signal (derived in Section V as the resolution-limited coupling), is the proof of solvency. The vacuum persists because its resolution limit is exactly minimally sufficient.

J. Conclusion: E_8-Persistence theory

The constraints of **Finiteness**, **Unitarity**, and **Causality** converge on a **unique** geometric structure: the E_8 lattice projected onto a causal $D = 4$ manifold accompanied by the set of Characteristic Integers: $\mathbb{S} = \{\Delta{=}43, \nu{=}16, \sigma{=}5, \chi{=}2\}$. Together these established a substrate on which other systems can persist.

V. SYSTEM 2: THE GEOMETRIC IMPEDANCE (α^{-1}), THE COST OF SUSTAINING TOPOLOGICAL CHARGE AGAINST THE LATTICE FLUX

At the substrate's fundamental resolution floor with zero external flux, the Entropic Action reduces to a static discrete sum over a single fundamental cycle ($\tau = 1$). For a persistent topological defect, this counts the minimum elementary operations required to sustain its configuration against the substrate's background stochastic fluctuations:

$$Z_{geo} = \frac{S_{\Phi,\text{defect}}}{Q_{top}} \tag{15}$$

where $S_{\Phi,\text{defect}}$ is the localized entropic action of the defect, and $Q_{top} := \chi/2 = 1$ is the unit topological charge (Section IV E 1). On a discrete lattice, this action is dimensionless: it counts lattice updates, link traversals, and symmetry transformations.[2]

Charge is not a postulated particle property but an emergent quantization of information flux. The topological boundary $\chi = 2$ (Section IV E 1) enforces a binary partition: information is either inside the defect or outside it. This $U(1)$ boundary invariant quantizes the flux through the defect. The unit charge is $Q_{top} = 1$.

Because the substrate is the only persistent system with no external reference, the strictly dimensionless information-theoretic resistance it presents to sustaining this unit charge is the only available measure of coupling strength. This pure number, defined here as the geometric impedance Z_{geo}, therefore governs how strongly a unit of topological charge couples to the lattice dynamics.

The Fine-Structure Constant ($\alpha^{-1} \approx 137$) is one of the most precisely measured couplings in physics. Standard quantum field theory treats it as an empirically measured parameter; its specific magnitude lacks a first-principles derivation. In standard quantum field theory, this dimensionless coupling value is parameterized in SI units as $\alpha = e^2/(4\pi\epsilon_0\hbar c)$. Here the geometric impedance Z_{geo} is identified with the inverse baseline coupling $\alpha^{-1} \equiv Z_{geo}$

Scale-dependent dynamics (running couplings and renormalization) must conform to this geometric baseline. The companion paper demonstrates that minimizing the Entropic Action on the E_8 lattice yields dynamics that remain strictly within these bounds, recovering the Standard Model and General Relativity as effective field theories.

A. The Geometric Impedance Equation

The valid functional form of Z_{geo} is uniquely determined by the Entropic Balance Sheet (section II B 3). Each term will be derived from the Characteristic Integers with zero empirical parameters; the sole non-integer coefficient, π, is the unique topological constant. Additivity of terms is guaranteed by strict architectural limits of E_8 Persistence Theory: exhausting the independent invariants of the $E_8 \to D_4 \oplus D_4$ projection and forbidding cross-terms via geometric orthogonality.

$$\begin{aligned} \alpha^{-1} \equiv Z_{geo} = \underbrace{\pi\Delta}_{CAP} + \underbrace{\chi}_{MAP} - \underbrace{\frac{1}{R_M - \sigma}}_{PRO} - \underbrace{\frac{\chi}{\Delta}}_{GOV} \\ + \underbrace{\frac{1 \cdot \chi \cdot (R_M - \sigma)}{N^3 \cdot \sigma \cdot R_M}}_{TOL} + \underbrace{\frac{1}{L_{embed} \cdot (\sigma + 1) \cdot \Delta^2}}_{MAR} \end{aligned} \tag{16}$$

B. The Fundamental Causal Circuit

The dominant contribution to the vacuum impedance ($\approx 99.9\%$) comes from the fundamental geometry of the interaction circuit. For a topological defect to persist in the lattice, it must complete a closed geometric cycle; the fundamental cycle Δ is the **resonant period** at which this closure occurs. We derive the impedance of this loop as the sum of the **Resonant Circumference** and the **Topological Boundary**. In dynamical gauge theory, the physical manifestation of this closed path is the Wilson Loop.

[2] An intuitive analogy: a communications channel has a maximum data rate set by its bandwidth and noise floor. For a signal to propagate, it must match the channel's impedance and exceed the noise floor. The Geometric Impedance is the total set of structural requirements a signal must meet to propagate losslessly on the physical substrate.

C. The Resonant Circumference (Capacity)

Form: The boundary action of the closed causal envelope.

Capacity measures the entropic footprint required for a topological defect to maintain structural coherence across its fundamental cycle. Without this cost the state would propagate openly at the bandwidth limit ($c = 1$) and delocalize completely.

Derivation: At Quiescent Equilibrium, the substrate's footprint is evaluated on its native positive-definite (Euclidean) metric required by Unitarity (section III C 1), with causal propagation restricted to a $(1+1)D$ plane (x, τ).

The fundamental cycle is periodic: the defect's configuration must recur identically after Δ updates or it does not persist. For any closed trajectory, reaching a distance d from the anchor costs at least d steps outbound and d steps to return, so $2d \leq \Delta$: the maximal one-way reach is $d_{\max} = \Delta/2$, saturated by the out-and-back path.

By Homogeneity the substrate admits no preferred direction, and saturating out-and-back paths exist along every orientation. The invariant envelope (the union of all closed Δ-cycles anchored at the node) is therefore the disk of radius $\Delta/2$: its diameter, the separation of two saturating excursions in opposite directions ($\Delta/2+\Delta/2$), is exactly Δ, matching the temporal period.

The defect is defined by its boundary (section IV E 1); the envelope's interior is indistinguishable from the vacuum, and impedance, as a substrate invariant, cannot depend on any particular trajectory through it. The entropic cost is therefore the boundary measure of the envelope, re-traced once per fundamental cycle. Because Homogeneity strictly forbids any preferred direction or lattice alignment, this invariant causal boundary must possess maximal rotational symmetry. The ratio of the isotropic boundary (the circle) to its diameter (Δ) introduces π is the geometric multiplier:

$$Z_{CAP} = \pi\Delta \approx 135.08848 \tag{17}$$

D. Topological Boundary (Map)

Form: The integer topological invariant of the boundary constraint.

The Map is the boundary constraint required to prevent internal state data from dissolving into the background substrate. Without this topological cost, the substrate could not distinguish the system's inside from its outside, causing information to leak into the unquantized background.

Derivation: As established in section IV E by the Gauss-Bonnet theorem, any closed, regular surface must satisfy a topological boundary condition. To prevent boundaries with holes ($g \geq 1$), where non-contractible loops destroy the strict binary partition of 'Inside' versus 'Outside' and permit unquantized information flux, the boundary must be simply-connected. The unique simply-connected, closed 2D surface is the sphere (S^2), which is characterized by the unique maximum Euler characteristic:

$$Z_{MAP} = +\chi = 2 \tag{18}$$

Because χ, the Euler characteristic is a pure topological invariant, it is an exact, indivisible integer. It is structurally forbidden from scaling, running, or acquiring continuous corrections.

E. Alignment Efficiency (Protocol)

Form: The negative reciprocal of the residual coordination capacity.

The Protocol mediates the flow of phase information across the orthogonal spacetime (Sector A) and internal symmetry (Sector B) channels. Without this alignment cost, these channels would completely decouple, fragmenting the vacuum into isolated, unrelated subspaces.

Derivation: The manifold resolution $R_M = D\Delta = 172$ is the total available coordination capacity, the maximum distinct spacetime addresses per fundamental cycle. A localized defect consumes exactly $\sigma = 5$ degrees of freedom to define its geometric embedding, leaving residual capacity:

$$C_{res} = R_M - \sigma = (4 \cdot 43) - 5 = 167 \tag{19}$$

Entropic impedance is the inverse of flow capacity. The greater the residual capacity, the lower the resistance to phase transport and because efficient coordination reduces net dissipation, this contribution is an organizational gain, strictly negative per the Entropic Balance Sheet (section II B 3):

$$Z_{PRO} = -\frac{1}{C_{res}} = -\frac{1}{167} \approx -0.00599 \tag{20}$$

The linear form is mandated by statelessness: the Protocol coordinates instantaneously, with no recursive memory that would permit higher-order couplings.

F. Stabilizing Potential (Governor)

Form: The negative ratio of the topological boundary to the causal cycle.

The Governor prevents runaway entropic divergence by constraining the resonant excitation across the temporal cycle. Without this stabilizing cost, local phase excitations on the causal loop would undergo unbounded, runaway feedback, overflowing the local node capacity.

Derivation: To satisfy the Homogeneity constraint, this boundary constraint cannot be anchored to any preferred temporal coordinate; it must be distributed maximally uniformly across the fundamental causal loop of length $\Delta = 43$.

Because the loop is closed and periodic, it possesses no endpoint boundaries (no "fencepost" corrections). To prevent non-linear self-interactions which would violate the vacuum's minimal interaction rule (the Toll constraint), the geometric coupling must be strictly first-order (linear). The exact stabilizing potential density per step is therefore the linear ratio of the boundary ($\chi = 2$) to the loop length ($\Delta = 43$). As an organizational gain that prevents entropic divergence, its contribution is negative:

$$Z_{GOV} = -\frac{\chi}{\Delta} = -\frac{2}{43} \approx -0.04651 \tag{21}$$

G. Entropic Transition (Toll)

Form: The ratio of persistence-preserving states to total interaction states.

The Toll is the entropic cost of executing a state transition. Without this update cost, the network's logical transitions would execute without a sequential causal history.

In geometric terms, an operational state transition fundamentally requires three components: an input, an output, and a mediating channel. On the substrate, an interaction is an operation on a node's channel space, not a statement about its geometric neighbors. By Schur's lemma, invariant bilinear couplings exist only within a single representation: a state can be propagated or measured (two legs), but never combined with a state of another type. The minimal invariant coupling of *distinct* channel types is the unique trilinear singlet of Spin(8) triality (equivalently, Clifford multiplication):

$$\mathbf{8}_v \otimes \mathbf{8}_s \otimes \mathbf{8}_c \longrightarrow \mathbf{1}.$$

Its three legs naturally assume the roles of a logic gate: input chirality, output chirality, and the mediating link channel. Because triality exhausts the 8-dimensional representations of Spin(8), a fourth leg would have to duplicate a type already present at the vertex; within one update cycle the node lacks the temporal resolution to distinguish duplicated sectors, producing the Causal Aliasing described in section III D. Higher-point transitions are therefore composite, resolvable only as sequences of 3-point vertices.

Derivation: A 3-point vertex has three independent legs. The denominator of the Toll is the *a priori* configuration space of the vertex: by the Closure Constraint, no external table pre-assigns channels to vertex roles, so each leg is unconstrained over the node's $N = 32$ channels, and by Homogeneity the three selections are independent. The total unconstrained configuration space is therefore $N \cdot N \cdot N = N^3$.

The total phase space Ω_{total} for a local 3-point event is the product of three orthogonal combinatorial factors: the N^3 channel combinations, the $\sigma = 5$ active interaction degrees of freedom, and the R_M manifold coordinate addresses:

$$\Omega_{total} = N^3 \cdot \sigma \cdot R_M \tag{22}$$

The valid persistence subset is defined by the intersection of three geometric constraints. Because these constraints operate on disjoint subspaces, the cardinality of their intersection is the exact product of their invariant limits:

1. **State Identity Projection (1):** The interaction must act as the identity operator on the state's internal channel configuration, yielding exactly 1 valid mapping to preserve state stability.
2. **Topological Flux Invariant ($\chi = 2$):** The interaction boundary is constrained by the topological manifold. The number of independent, non-vanishing flux orientations through this closed boundary is defined strictly by its Euler characteristic ($\chi = 2$).
3. **Coordinate Non-Degeneracy ($R_M - \sigma$):** To avoid causal aliasing, the transition target cannot project onto the σ coordinates already occupied by the interaction's own geometric embedding. The valid output is strictly restricted to the orthogonal residual subspace, which has exactly $(R_M - \sigma)$ coordinates.

$$\Omega_{valid} = 1 \cdot \chi \cdot (R_M - \sigma) \tag{23}$$

The entropic cost (Toll) of a state transition is exactly the phase-space ratio of finding a valid persistence configuration within the total interaction space:

$$Z_{TOL} = \frac{\Omega_{valid}}{\Omega_{total}} = \frac{1 \cdot \chi \cdot (R_M - \sigma)}{N^3 \cdot \sigma \cdot R_M} \approx 1.1852 \cdot 10^{-5} \tag{24}$$

H. Resolution Floor (Margin)

Form: The reciprocal of the orthogonal configuration phase-space volume.

At the substrate level, the safety buffer manifests as an information-theoretic distinguishability threshold: the minimum signal amplitude required to be resolved above the geometric noise floor, analogous to the gain margin that prevents control-loop instability. Below this floor, structured information dissolves into ambient substrate fluctuations.

Derivation: The minimum resolvable signal is exactly 1 quantum of information diluted across the total configuration volume (V_{config}). Because the $E_8 \to D_4 \oplus D_4$

projection partitions the lattice into strictly orthogonal sectors, these sectors are statistically independent, meaning their configuration volumes combine via their Cartesian product.

This total joint configuration phase space is the product of three independent, mathematically forced factors:

$$V_{config} = L_{embed} \cdot (\sigma + 1) \cdot \Delta^2 \tag{25}$$

where:

1. **The State Vector ($L_{embed} = 31$):** The total description length of a dynamic defect. Because its defining properties are strictly orthogonal information channels—internal identity ($\nu = 16$, $\chi = 2$), geometric anchor ($\sigma = 5$), and kinematic phase space ($2D = 8$, specifying position and transition rate), their data capacities combine via direct sum: $L_{embed} = \nu + \sigma + \chi + 2D = 31$.

2. **The Interaction Aperture ($(\sigma + 1) = 6$):** The exact cardinality of the active closed neighborhood on the discrete lattice. A localized interaction operator must span the $\sigma = 5$ active links mediating the flow, plus exactly 1 central vertex hosting the state's occupancy.

3. **The Causal Area (Δ^2):** Because causality restricts information propagation to a $(1+1)$D plane at the maximum speed limit $c \equiv 1$, a fundamental cycle of temporal duration Δ sweeps a spatial horizon of length Δ. The resulting macroscopic spacetime area is strictly $\Delta \cdot \Delta = \Delta^2$.

The Margin impedance is the reciprocal, the exact informational resolution limit of a single localized quantum:

$$\begin{aligned} Z_{MAR} &= \frac{1}{V_{config}} = \frac{1}{L_{embed} \cdot (\sigma + 1) \cdot \Delta^2} \\ &= \frac{1}{31 \cdot (5+1) \cdot 43^2} \approx 2.9077 \cdot 10^{-6} \end{aligned} \tag{26}$$

I. Substrate Orthogonality and the Additive Baseline

While macroscopic control systems exhibit non-linear cross-coupling due to shared, continuous state spaces, the vacuum substrate at the absolute resolution floor (Quiescent Equilibrium) strictly diagonalizes the lowest-order action. Because the six pillars operate on geometrically orthogonal subspaces within the $E_8 \rightarrow D_4 \oplus D_4$ projection, their existential constraints are statistically independent. Consequently, by the principles of algorithmic information theory, their joint entropic impedance is strictly additive:

$$Z_{geo} = \sum_{i=1}^{6} Z_i \tag{27}$$

These degrees of freedom map to disjoint geometric domains: Capacity and Margin to metric volume, Map to topological boundaries, Governor to temporal constraint density, and Toll to combinatorial channel selection. The interaction term Z_{PRO} mediates between the $D_4 \oplus D_4$ sectors by operating on the residual coordination capacity $(R_M - \sigma)$ that spans the boundary, quantifying the interface alignment cost without violating sector disjointness.

Truncation of Non-Linear Cross-Terms: Macroscopic network theory would predict higher-order recursive loops to generate non-linear cross-terms in this summation. In this framework, such recursive interactions are strictly constrained: they manifest dynamically as the scale-dependent running of the coupling constant (renormalization), but must remain within the bounds of the bare geometric impedance derived here. The companion paper demonstrates that renormalization group flows on this substrate naturally respect this resolution floor.

The value derived herein is exclusively the *bare* geometric impedance—the atomic, lowest-order resolution limit of the lattice. At this indivisible structural floor, operations are discrete and minimal. Because the substrate lacks the local hardware capacity to resolve sub-steps beneath the fundamental cycle Δ, sub-step interference cross-terms are physically forbidden by finite bit-depth, rigidly locking the baseline evaluation to strict linearity.

J. Completeness and the Finite Sum

Two constraints rigidly bound the derivation.

First, the six pillars exhaust the architectural degrees of freedom required for persistence; each term corresponds one-to-one with a component of the Universal Architecture. The functional form of each term is established individually in its derivation (sections V C to V H), forced by the geometric and information-theoretic requirements of its pillar. The signs are fixed by the Entropic Balance Sheet (section II B 3). Terms representing irreducible structural overhead (Capacity, Map, Toll, Margin) are positive; terms representing organizational gains (Protocol, Governor) are negative. This classification is established in the theoretical framework and confirmed by the failure mode analysis (section II C).

Second, the characteristic integers $\mathbb{S} = \{\Delta = 43, \nu = 16, \sigma = 5, \chi = 2\}$ exhaust the independent invariants of the $\mathrm{E}_8 \rightarrow D_4 \oplus D_4$ projection. Note that $\nu = 16$ enters only through $N = 2\nu = 32$; no term involves ν independently because the chiral bit-depth is fully allocated to the channel structure (section IV C). No additional independent invariants have been identified. Cross-terms are forbidden because the pillars operate on orthogonal subspaces (section V I).

The summation $\alpha^{-1} = \sum_{i=1}^{6} Z_i$ is therefore the minimal and complete lowest-order persistence basis consistent with the substrate geometry.

K. Numerical Validation

Summing the geometric components:

$$\alpha_{calc}^{-1} = 137.035999212\ldots \tag{28}$$

Because π is the sole non-integer constant, all other terms being exact rational combinations of the characteristic integers, the precision of the prediction is limited only by the evaluation of π.

Recent experimental values (Table I) do not form a simple consensus and reflect ongoing metrological tension. Morel (2020) and Parker (2018) measure α^{-1} kinematically using photon recoil (Rubidium and Cesium, respectively). Fan (2023) determines α^{-1} via the electron anomalous magnetic moment $(g-2)$. That these independent measurements are mutually exclusive is an open problem in standard metrology. CODATA (2022) synthesizes these conflicting inputs and its uncertainty reflects this tension.

Source	**Value** (α^{-1})	**Dev.**
Morel (2020)	$137.035\,999\,206 \pm 0.000\,000\,011$ [21]	0.58σ
Fan (2023)	$137.035\,999\,166 \pm 0.000\,000\,015$ [22]	3.09σ
Parker (2018)	$137.035\,999\,046 \pm 0.000\,000\,027$ [23]	6.16σ
CODATA (2022)	$137.035\,999\,177 \pm 0.000\,000\,021$ [24]	1.68σ

Table I. Comparison of the E_8-Persistence Theory α^{-1} derivation against experimental values. Deviation is computed as $|\alpha_{calc}^{-1} - \alpha_{exp}^{-1}|/\sigma_{exp}$.

a. The QED Extraction Tension and the Quantization Gap: E_8-Persistence predicts the *Quantization Gap* between kinematic measurements and high-order perturbative QED extractions when loop complexity approaches the substrate's finite channel capacity. The 3.09σ deviation observed in Fan (2023) is suggestively aligned with this predicted threshold.

Morel (2020) measures α kinematically. This would be a direct macroscopic probe of the geometric impedance that does not rely on quantum loop corrections. Fan (2023), however, extracts α by fitting experimental data to a 10th-order (5-loop) QED perturbative expansion.

Standard QED calculations are the analytic dual to this discrete architecture. They utilize dimensional regularization, which maps to the low-capacity limit of the substrate by treating spacetime as a continuous manifold with infinite resolution. At lower loop orders, this continuous approximation is extraordinarily successful. However, at 5-loop QED, the calculation achieves a fractional precision of $\sim (\alpha/\pi)^5 \sim 10^{-10}$. At this extreme threshold, the continuous-spacetime dual attempts to resolve state configurations that exceed the Shannon channel capacity of the underlying discrete hardware.

We term this $\sim 3 \times 10^{-10}$ threshold the **Quantization Gap**. This resonates with Freeman Dyson's 1952 observation[25] that the QED perturbative series is asymptotic rather than convergent. Our framework provides a structural mechanism for this behavior: it acts as the information-theoretic limit where a continuous mathematical approximation attempts to resolve state configurations that exceed the Shannon channel capacity of the discrete underlying hardware.

This breakdown scale can be heuristically estimated by comparing Feynman diagram multiplicity to the substrate's channel capacity. The number of QED diagrams grows factorially with loop order; at 5-loop precision, the electron anomalous magnetic moment calculation requires evaluating 12,672 distinct diagrams. Simultaneously, the available combinatorial state space of the fundamental causal loop scales as $\nu^n = 16^n$. At $n = 4$ to $n = 5$, the factorial diagram complexity ($\sim 10^4$) reaches parity with the hardware phase-space limit ($16^4 = 65,536$), marking the threshold where the continuous expansion saturates the substrate's discrete bit-depth.

Falsification criterion: If pure kinematic measurements continue to converge to $\alpha^{-1} \approx 137.035999212$ while 6+ loop QED extractions systematically diverge from this value, the Quantization Gap is confirmed. Conversely, the convergence of both methods to a single, identical value differing from our geometric prediction by $> 5\sigma$ would decisively falsify the E_8-Persistence theory.

L. Corollary: Charge from Geometric Impedance

Because the geometric impedance Z_{geo} is identified with α^{-1}, the standard electrodynamic definitions of e and R_K become consistency constraints on the substrate. The elementary charge e is not an input but an output: it is the information flux that the substrate permits through a $\chi = 2$ topological boundary when the impedance is Z_{geo}.

By the standard definition $\alpha = e^2/(4\pi\epsilon_0\hbar c)$, the elementary charge is:

$$e = \sqrt{\frac{4\pi\epsilon_0\hbar c}{Z_{geo}}} \tag{29}$$

Charge is therefore a flow constraint: the geometric impedance restricts information flux through a $\chi = 2$ topological defect to discrete values. Quantization follows from integer invariants; no additional gauge-theoretic mechanism is required.

M. The Von Klitzing Constant (R_K)

While the relation $R_K = Z_0/2\alpha$ is a known algebraic identity in standard electromagnetism, the E_8-Persistence theory provides a novel structural interpretation for its components. Specifically, the factor of 2 emerges geometrically from the Spinor Double Cover; it corresponds to the winding number required for a spinor to return to its initial phase after traversing the manifold.

A geometric impedance should manifest as a measurable physical resistance. The Quantum Hall Effect provides exactly this test: it measures the resistance of a single quantum channel with parts-per-billion precision, independently of α^{-1}. Three structural facts of the substrate yield R_K directly:

1. **Single Channel Limit:** The vacuum substrate supports discrete transmission channels. One channel has impedance $Z_{channel}$.
2. **Spinor Double Cover:** Fermions are spinors. To complete a closed circuit and return to initial phase, the carrier traverses the manifold twice (720°). The measured resistance is the vacuum impedance shared across two windings: $R_K = Z_{channel}/2$.
3. **Geometric Coupling:** The Characteristic Impedance of Free Space Z_0 is the macroscopic transmission resistance of the lattice substrate itself. The framework derives the dimensionless scaling between a single channel and the bulk medium as $Z_{channel}/Z_0 = \alpha_{geo}^{-1}$.

Combining these yields the strict geometric relationship $R_K/Z_0 = \alpha_{geo}^{-1}/2$. We evaluate it in the SI unit system by applying the macroscopic vacuum impedance isomorphism ($Z_0 = \mu_0 c \approx 376.73\,\Omega$) to compare our dimensionless geometric ratio with laboratory measurements:

$$R_K = \frac{Z_0}{2} \cdot \alpha_{geo}^{-1} \approx 25\,812.807\,469\,41\,\Omega \tag{30}$$

- **Experimental Value (CODATA 2022):** $25\,812.807\,45 \pm 0.000\,01\,\Omega$ [24]
- **Precision:** The geometric prediction lies within 1.94σ of the Quantum Hall resistance.

Physical consequence: The flatness of QHE plateaus. $R = R_K/n$, regardless of impurities, reflects the lattice's discrete bit-depth: resistance is quantized by channel count, offering a structural origin for the topological protection traditionally attributed to electron wavefunction topology (Chern numbers) [26, 27].

N. The Planck Charge Ratio

The ratio $e/q_P \approx 0.085$ is a known numerical identity but has lacked structural explanation: why should the electromagnetic coupling set the attenuation between natural and observed charge? In the E_8-Persistence framework, this acts as a structural necessity: the geometric impedance Z_{geo} enforces a safety margin between the operating charge and substrate breakdown.

The elementary charge appears as the Planck charge attenuated by the square root of the lattice impedance:

$$e = \frac{q_P}{\sqrt{Z_{geo}}} \approx \frac{q_P}{11.7} \tag{31}$$

1. *Physical consequence: Safe Load Limit.*

The electron represents the **safe operating limit** of the vacuum. While the substrate can theoretically support a unitary charge (q_P), the geometric impedance restricts propagating charge to $\approx 8.5\%$ of this maximum.

2. *Physical consequence: Vacuum Breakdown as Capacity Limit (Schwinger Limit).*

The Schwinger critical field $E_c = m_e^2 c^3/e\hbar$ marks the threshold where external electric fields spontaneously produce e^+e^- pairs [28]. The Coulomb field of an elementary charge, evaluated at the reduced Compton wavelength $\lambda = \hbar/m_e c$, scales with this critical value by exactly the fine-structure constant α:

$$\frac{e}{4\pi\epsilon_0 \lambda^2} = \alpha E_c = \frac{E_c}{Z_{geo}} \tag{32}$$

The dimensionless geometric impedance factor $Z_{geo} \approx 137$ therefore represents the **safety factor** between the characteristic field of an isolated fundamental charge and the vacuum breakdown threshold. Any localized excitation attempting to concentrate flux beyond this margin hits the non-linear Schwinger ceiling, resolving into particle-antiparticle pairs and enforcing the substrate's ultimate capacity limit.

VI. CONCLUSION

This work introduced *Informational Energetics*, a framework deriving the minimal architecture of persistence from control theory, information theory, and thermodynamics. We established that any system resisting entropic decay requires six irreducible components (Capacity, Map, Protocol, Governor, Toll, Margin), each demonstrated by its catastrophic failure mode when removed.

We tested IE in its most demanding domain: fundamental physics. We proposed that the fundamental structures of reality are the emergent consequences of a universe optimized for persistence. Translating the six pillars to physical requirements (Finiteness, Unitarity, Causality) strictly constrained the viable vacuum substrate to the E_8 lattice projected onto a causal 4D manifold. This projection intrinsically generates the Lorentzian metric signature, operating outside the assumptions of the Distler-Garibaldi no-go theorem, yielding the Characteristic Integers $\mathbb{S} = \{\Delta = 43, \nu = 16, \sigma = 5, \chi = 2\}$ with no free parameters.

The derivation was a sequence of constraints isolating the optimal substrate, not an open search over possibilities. This provides a stark contrast to predominant unification programs. Frameworks such as 10D Superstring Theory, in their current formulations, yield a vast landscape of potential vacua, requiring external compactification manifolds and auxiliary parameters to match observable reality. The Closure Constraint demands a strictly self-defining substrate requiring zero bits of external selection information. The dimensional constraint $n \equiv 0 \pmod 8$ then dictates that the observed 4D spacetime is an emergent projection of this optimal hardware. The fine-structure constant emerges as the unavoidable thermodynamic tax required to maintain structural coherence on this specific substrate.

The ultimate test of this static architecture is falsifiable contact with experiment. The geometric impedance Z_{geo} yields a parameter-free prediction for the fine-structure constant ($\alpha^{-1} = 137.035999212$). This value structurally aligns with the kinematic recoil measurement of Morel (2020) at 0.58σ and CODATA (2022) at 1.68σ, while predicting a strict *Quantization Gap* where continuous, high-loop QED extractions must systematically diverge due to the substrate's finite channel capacity. The same impedance determines R_K to 0.08 parts per billion and the elementary charge $e = q_P/\sqrt{Z_{geo}}$, which emerges strictly as a flow constraint enforcing the Schwinger vacuum breakdown limit. The Bohr radius emerges as $a_0 \propto Z_{geo}/Z_{MAR}$, locking the scale of chemistry to the lattice resolution.

The IE architecture established here defines the strict boundary conditions that any valid physical Lagrangian must satisfy. The companion paper *Informational Energetics: Entropic Action* formalizes the Entropic Action and demonstrates that minimized the action on this E_8 substrate, naturally yields dynamics that remain within these exact bounds, recovering the Standard Model and General Relativity as effective field theories. This cross-validation is strict: the static substrate derived here imposes boundary conditions that any valid dynamics must respect; the companion paper demonstrates that thermodynamic minimization on this substrate naturally yields operators satisfying these conditions, converging on the exact same physics.

These boundary conditions further constrain how systems must partition themselves when internal complexity exceeds the capacity of a single control loop. The companion paper*Informational Energetics: Nested Persistence* derives the universal partitioning rules consistent with these constraints. While applying these rules to the vacuum derives the remaining Standard Model constants or recasting the CKM/PMNS matrices as the minimal entropic cost of multi-state routing these exact same partitioning mechanics apply to the hierarchical scaling of biological metabolisms and economic networks.

The derivation of the vacuum substrate and its geometric impedance is a proof of concept, not the endpoint. Informational Energetics proposes that persistence is the primary organizing principle of reality at every scale. While macroscopic entities from biological cells to global economic networks must navigate their existential constraints probabilistically against chaotic environments, the vacuum serves as the absolute, zero-flux baseline.

If the fundamental vacuum is uniquely determined by the strict constraints of persistence, then by the Selection Principle, any persistent macroscopic system is solving the exact same architectural problem, differing only in substrate material and environmental openness. Ultimately, the fabric of physical reality is not the foundation of this framework; it is merely its cleanest, most rigorous first test. The static geometric impedance derived herein establishes the immutable kinematic floor—the mandatory boundary conditions—upon which all subsequent dynamic, macroscopic, and entropic actions must be built.

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[1] R. Landauer, Irreversibility and Heat Generation in the Computing Process, IBM Journal of Research and Development **5**, 183 (1961).

[2] J. A. Wheeler, Information, Physics, Quantum: The Search for Links, in *Proceedings III International Symposium on Foundations of Quantum Mechanics*, edited by W. J. Archibald (Physical Society of Japan, 1989) pp. 354–358.

[3] E. Verlinde, On the origin of gravity and the laws of Newton, Journal of High Energy Physics **2011**, 29 (2011).

[4] G. Bianconi, The Thermodynamics of the Gravity from Entropy Theory (2025), version Number: 2.

[5] A. G. Lisi, An Exceptionally Simple Theory of Everything (2007), arXiv:0711.0770 [hep-th].

[6] J. Distler and S. Garibaldi, There is No "Theory of Everything" Inside E8, Communications in Mathematical Physics **298**, 419 (2010).

[7] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, Spacetime as a causal set, Physical Review Letters **59**, 521 (1987).

[8] C. Rovelli, *Quantum Gravity*, 1st ed. (Cambridge University Press, 2004).

[9] T. Jacobson, Thermodynamics of Spacetime: The Einstein Equation of State, Physical Review Letters **75**, 1260 (1995).

[10] R. C. Dorf and R. H. Bishop, *Modern Control Systems*, 13th ed. (Pearson, 2017).

[11] K. J. Åström and R. M. Murray, *Feedback systems: an*

introduction for scientists and engineers (Princeton University Press, Princeton, 2008).
[12] N. Margolus and L. B. Levitin, The maximum speed of dynamical evolution, Physica D: Nonlinear Phenomena **120**, 188 (1998).
[13] J. D. Bekenstein, Universal upper bound on the entropy-to-energy ratio for bounded systems, Physical Review D **23**, 287 (1981).
[14] R. C. Conant and W. Ross Ashby, Every good regulator of a system must be a model of that system †, International Journal of Systems Science **1**, 89 (1970).
[15] C. Shannon, Communication in the Presence of Noise, Proceedings of the IRE **37**, 10 (1949).
[16] J. D. R. D. Raadt, ASHBY'S LAW OF REQUISITE VARIETY: AN EMPIRICAL STUDY, Cybernetics and Systems **18**, 517 (1987).
[17] H. Nyquist, Regeneration Theory, Bell System Technical Journal **11**, 126 (1932).
[18] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, edited by M. Artin, S. S. Chern, J. M. Fröhlich, E. Heinz, H. Hironaka, F. Hirzebruch, L. Hörmander, S. Mac Lane, C. C. Moore, J. K. Moser, M. Nagata, W. Schmidt, D. S. Scott, Y. G. Sinai, J. Tits, B. L. Van Der Waerden, M. Waldschmidt, S. Watanabe, M. Berger, B. Eckmann, and S. R. S. Varadhan, Grundlehren der mathematischen Wissenschaften, Vol. 290 (Springer New York, New York, NY, 1988).
[19] M. Viazovska, The sphere packing problem in dimension 8, Annals of Mathematics **185**, 10.4007/annals.2017.185.3.7 (2017).
[20] K. Immink, Runlength-limited sequences, Proceedings of the IEEE **78**, 1745 (1990).
[21] L. Morel, Z. Yao, P. Cladé, and S. Guellati-Khélifa, Determination of the fine-structure constant with an accuracy of 81 parts per trillion, Nature **588**, 61 (2020).
[22] X. Fan, T. Myers, B. Sukra, and G. Gabrielse, Measurement of the Electron Magnetic Moment, Physical Review Letters **130**, 071801 (2023).
[23] R. H. Parker, C. Yu, W. Zhong, B. Estey, and H. Müller, Measurement of the fine-structure constant as a test of the Standard Model, Science **360**, 191 (2018).
[24] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, CODATA recommended values of the fundamental physical constants: 2022, Reviews of Modern Physics **97**, 025002 (2025).
[25] F. J. Dyson, Divergence of Perturbation Theory in Quantum Electrodynamics, Physical Review **85**, 631 (1952).
[26] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. Den Nijs, Quantized Hall Conductance in a Two-Dimensional Periodic Potential, Physical Review Letters **49**, 405 (1982).
[27] M. Kohmoto, Topological invariant and the quantization of the Hall conductance, Annals of Physics **160**, 343 (1985).
[28] J. Schwinger, On Gauge Invariance and Vacuum Polarization, Physical Review **82**, 664 (1951).

Appendix A: Geometric Origin of Chiral Fermions

Distler and Garibaldi [6] demonstrated that a signature-preserving embedding of the Standard Model within E_8 cannot accommodate chiral fermions without unobserved mirror partners. Their theorem holds for all embeddings where the algebraic structure remains Euclidean $(8, 0)$. We include this brief derivation for readers seeking explicit verification that the metric signature change $(8, 0) \to (1, 3) \oplus (4, 0)$ evades the theorem by rendering mirror states dynamically inert.

The E_8-Persistence theory employs the symmetric decomposition $E_8 \to D_4 \oplus D_4$ (derived in section IV B 1) coupled with the Metric Projection $(8, 0) \to (1, 3) \oplus (4, 0)$ (derived in section IV D), which falls outside this assumption. As established in section IV D, the internal symmetry sector retains the original Euclidean signature $(4, 0)$ and therefore lacks a timelike generator.

a. Formal Resolution: The Distler-Garibaldi constraint assumes a *signature-preserving continuous Lie algebra* embedding. We explicitly violate this assumption. Because the E_8 lattice is a discrete geometric object, rather than a continuous algebraic homomorphism, the Causality requirement strictly forces a geometric subspace restriction onto an observable Lorentzian spacetime manifold (e.g., Minkowski space $\mathbb{R}^{1,3}$). The orthogonal degrees of freedom, which contain the mirror states, reside strictly within the Euclidean internal sector.

Because this Euclidean sector possesses signature $(+, +, +, +)$, its isometry group is the compact rotation group $SO(4)$. This group fundamentally lacks the timelike translation generator ($P_0 = \partial_t$) present in the spacetime Poincaré group $ISO(1, 3)$ required for dynamical evolution.

Consequently, because the internal $SO(4)$ group and the spacetime Poincaré group commute, the mirror states ψ_{mirror} possess no temporal Hamiltonian relative to the observable manifold. They satisfy the strictly stationary condition $\partial_t \psi_{\text{mirror}} = 0$. In an information-theoretic framework, a state without a time-evolution operator cannot propagate or encode a changing signal; it acts entirely as a static internal degree of freedom (a quantum number). Mirror fermions are therefore **geometrically suppressed**. They do not propagate on the observable $(1, 3)$ manifold, evading the Distler-Garibaldi theorem by violating its implicit premise: that the full E_8 algebra acts dynamically in spacetime.

While this geometric suppression evades the no-go theorem at the level of the static substrate, the dynamical decoupling of these mirror states, and the geometric evasion of lattice fermion doubling, is formally enforced by the chiral projection operators. The rigorous derivation of this chiral filter, demonstrating that mirror states are rendered as static singlets under the temporal flow, is deferred to the companion paper on *Informational Energetics: Entropic Action*, where the dynamical Lagrangian is derived.

Appendix B: Algebraic Geometry of the Causal Loop and Complex Multiplication

This appendix provides the formal mathematical derivation bridging the physical requirement of Unitarity to the algebraic necessity of the Stark-Heegner numbers, as referenced in section IV F 1.

1. The Setup: Dimensional Reduction to the Causal Plane

Information propagation on the E_8 substrate ($D_4 \oplus D_4$) is fundamentally local: a signal from one node to another defines a 1-dimensional spatial trajectory evolving over 1-dimensional time, restricting causal propagation to a $(1+1)D$ **Causal Plane**.

Because the vacuum possesses a fundamental temporal cycle, this sequence of updates must eventually close, forming a periodic loop of length Δ.

2. Mathematical Translation: The Elliptic Curve

Topologically, a discrete, periodic $(1+1)D$ causal plane forms a 2-dimensional real torus ($\mathbb{T}^2$). To analyze this rigorously as a 1-dimensional complex torus (an Elliptic Curve), it must be endowed with a complex structure.

The causal plane is spanned by one spatial step (δx) and one temporal step (δt). As derived in section IV D, the Lorentzian metric signature assigns the temporal dimension a geometric imaginary character: $\delta t = i \cdot \delta x$ (in natural units, $c = 1$).

Consequently, the causal lattice tiles the complex plane $\mathbb{C}$ with a real spatial period $\omega_1 = 1$ and a geometric imaginary temporal period. This defines a complex lattice $\Lambda = \mathbb{Z} \cdot 1 \oplus \mathbb{Z} \cdot z$, yielding the elliptic curve $\mathbb{C}/\Lambda$ governed strictly by the complex modular parameter z.

Because the trajectory is physically constrained to the discrete, self-dual structure of the underlying E_8 lattice, z cannot be a generic continuous complex variable. The discrete, self-dual lattice structure constrains z to a Complex Multiplication (CM) point, a fixed point of a Möbius transformation with integer coefficients acting on the moduli space. This requires z to satisfy a quadratic equation $az^2 + bz + c = 0$ with $a, b, c \in \mathbb{Z}$ and discriminant $b^2 - 4ac = -D < 0$.

3. Lemma 1: The Identification of Period and Discriminant ($D = \Delta$)

To utilize the theory of Complex Multiplication, we must formally bridge the physical geometric period Δ to the algebraic CM discriminant D.

Lemma: *For a unitary causal loop on a self-dual discrete lattice, the geometric temporal period Δ is identically equal to the CM discriminant D.*

Proof:

1. **Lattice Basis:** The causal loop defines a complex torus $\mathbb{C}/\Lambda$ with lattice basis $\{1, z\}$, where the real axis represents the spatial step and the imaginary axis represents the temporal step.

2. **The Geometric Norm:** The temporal period Δ dictates that the state history closes exactly after Δ discrete updates. Algebraically, this requires the temporal generator z to satisfy $z^2 + \Delta = 0$, placing the system in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-\Delta})$.

3. **Self-Duality Constraint:** To preserve the self-duality of the parent E_8 substrate ($\Lambda = \Lambda^*$), the partition function $Z_\Lambda(z) = \Theta_\Lambda(z)/\eta(z)^8$ must be invariant under the modular inversion $S : z \to -1/z$. This restricts z strictly to the ring of integers $\mathcal{O}_K$ of the field K.

4. **The Discriminant Mapping:** From Step 2, the causal loop resides in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-\Delta})$. By the theory of algebraic number fields, the fundamental discriminant of K equals $-\Delta$ when $\Delta \equiv 3 \pmod 4$, and -4Δ otherwise. Inspection of the Stark-Heegner candidates surviving the Causality constraint $\Delta > N = 32$, namely $\{43, 67, 163\}$, confirms that each satisfies $\Delta \equiv 3 \pmod 4$. Therefore, the fundamental discriminant of K is exactly $-\Delta$, and the magnitude of the CM discriminant $D = |\mathrm{disc}(K)| = \Delta$ exactly, with no fractional remainder. ■

4. The Failure Mode: Isogeny Classes and Causal Degeneracy

For the system to preserve quantum Unitarity, the history of a state traversing this causal loop must be uniquely reconstructable. Mathematically, this requires the endomorphism ring of the elliptic curve to be a Principal Ideal Domain (PID), which corresponds to a class number of $h = 1$.

We analyze the failure mode where $h > 1$. In this regime, the ideal class group of the field is non-trivial. By the theory of Complex Multiplication, this dictates that multiple non-isomorphic elliptic curves will share the exact same endomorphism ring.

Physically, these isogenous curves represent topologically distinct causal histories. However, because they are locally indistinguishable by their norm structure, a physical measurement at the output node cannot determine which of the h inequivalent curves generated the state.

This local indistinguishability manifests as **Causal Degeneracy**. The causal history is fundamentally erased, not merely unknown. This ambiguity explicitly violates the unitary requirement that the evolution operator be uniquely invertible ($\hat{U}^\dagger \hat{U} = I$).

5. The Resolution: The Unitarity Constraint

To prevent effective information loss and preserve Unitarity, the universe must strictly forbid non-trivial ideal class groups in its fundamental causal cycles. The algebraic field defining the cycle must possess $h = 1$. By Lemma 1 ($D = \Delta$), this strictly restricts the geometric temporal period Δ to the Stark-Heegner numbers $D \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$.

While the $h = 1$ constraint restricts Δ to the Stark-Heegner numbers, the final selection of $\Delta = 43$ is enforced by the Self-Clocking phase-lock limit derived in section IV F.